

Multisensor Gait Data Collection Pipeline with Spectrum-based EMG Reconstruction for Neural Network Models

URA Report, Spring 2025

Grace Kim (Undergraduate Research Assistant)
Mechanical and Mechatronics Engineering
University of Waterloo
Waterloo, Canada
g72kim@uwaterloo.ca

Ehsan Tahvilian (Supervisor)
Mechanical and Mechatronics Engineering
University of Waterloo
Waterloo, Canada
ehsan.tahvilian@uwaterloo.ca

Abstract—This project focused on collecting and preprocessing gait cycle data to support the future development of neural network models for rehabilitation. Human walking trials were captured from ten participants using a Vicon motion capture system, electromyography (EMG) sensors, inertial measurement units (IMU), and force plates. Preprocessing steps included marker gap filling, anatomical labelling, kinematic model fitting, IMU–Vicon synchronization, and artificial reconstruction of EMG disconnections. Challenges such as EMG dropout were addressed through signal processing and mathematical reconstruction methods involving interpolation, frequency-domain feature extraction, and noise modeling. The resulting high-quality, continuous datasets will be used for training neural networks designed to generate one future gait cycle of joint trajectories for exoskeleton-based rehabilitation applications.

Index Terms—Gait analysis, Motion capture, EMG, IMU, Synchronization, Spectrum-based Reconstruction, Signal processing, Neural networks.

I. INTRODUCTION

Neurological disorders such as stroke and spinal cord injury affect millions of people globally and often result in partial or complete loss of mobility [1], [2]. Walking, a seemingly effortless daily activity, is essential for independence, autonomy, and quality of life [3]. Impaired gait leads to significant physical and psychological challenges. Rehabilitation technologies—including robotic exoskeletons and machine-learning-based assistive systems—have shown promise in restoring motor function and supporting repetitive movement training [4].

This research project aims to build a structured and reproducible gait data pipeline as the foundation for future neural network models that predict human joint trajectories. As an Undergraduate Research Assistant (URA), the main objectives were:

- Collect gait cycle data using Vicon motion capture, EMG sensors, and IMUs.
- Preprocess the data through marker gap filling, anatomical labelling, synchronization, and filtering.
- Prepare the dataset for next term’s work involving neural network prediction of future gait cycles.

To support these goals, this report presents a coherent description of the full workflow and highlights practical lessons learned when working with human motion datasets.

II. BACKGROUND

Human gait is a periodic motion that can be modeled mathematically. A gait cycle is usually represented from heel strike to the next heel strike of the same side. Let

$$\mathbf{x}(t) \in \mathbb{R}^n$$

represent joint or marker kinematic variables at time t , where n is the number of kinematic degrees of freedom (e.g., marker coordinates or joint angles).

Gait signals often exhibit quasi-periodicity, and a neural network trained on a sequence

$$\{\mathbf{x}(t_1), \mathbf{x}(t_2), \dots, \mathbf{x}(t_k)\}$$

may be used to predict a future trajectory

$$\mathbf{x}(t_{k+1}), \dots, \mathbf{x}(t_{k+m}),$$

where k is the length of the observed input sequence and m is the number of future time steps to be predicted.

Accurate predictions depend critically on clean, aligned, and high-fidelity data. Noise, missing markers, EMG dropouts, and sensor time offsets degrade model performance if not handled rigorously. For this reason, robust preprocessing is a necessary component of gait-based machine-learning pipelines.

III. METHODS

In this section, the methodology is described in detail, from data collection and calibration to the full EMG preprocessing and automated correction logic. The goal is to provide a mathematically explicit formulation of the algorithms that were implemented in Python.

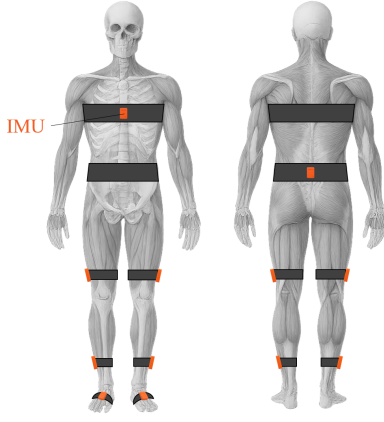


Fig. 1. IMU sensor attachment guidelines.

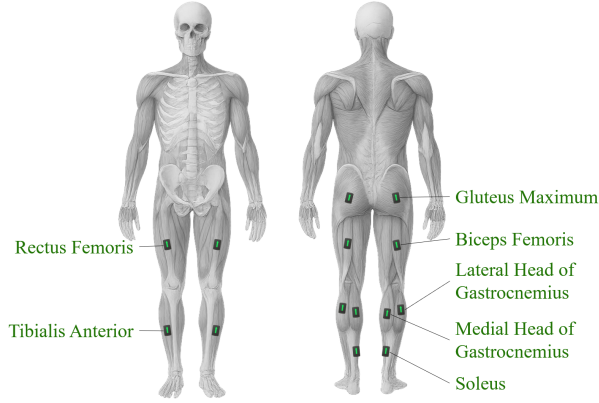


Fig. 2. EMG sensor attachment guidelines.

A. Experimental Setup

Data were collected using the following equipment:

- Vicon Motion Capture System (Nexus software, infrared cameras, passive reflective markers).
- Surface EMG sensors.
- IMU sensors attached to lower-limb segments.
- Instrumented treadmill and Bertec force plates.

Ten able-bodied participants (five male and five female) performed two walking protocols:

- 1) Six constant-speed trials at velocities ranging from 400–1400 mm/s, with increments of 200 mm/s every 90 s.
- 2) A continuous speed-ramping trial increasing from 400 to 1400 mm/s over 240 s, then decreasing back to 400 mm/s.

B. Calibration Procedures

A two-stage calibration procedure was used:

- **Static calibration** to establish anatomical landmarks and scaled body segments.
- **Functional calibration** to estimate joint axes through controlled movements.

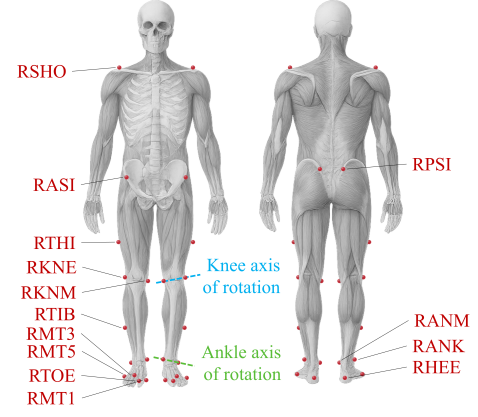


Fig. 3. Illustration of marker placement for defining joint axes.

TABLE I
GAP-FILLING METHODS IN VICON NEXUS.

Method	Description	Use Case
Spline fill	Cubic interpolation	Small gaps (< 10 frames)
Cyclic fill	Uses periodic gait patterns	Periodic missing segments
Rigid body fill	Reconstructs markers using rigid triplets	Hip/pelvis segments
Pattern fill	Follows reference marker pattern	Inner/outer knee markers
Kinematic fill	Uses model-based estimates	Rarely used

Starting from the third participant, four extra medial markers (LKNM, RKNM, LANM, RANM) were added to accurately capture the axes of rotation of the knee and ankle. Marker pairs defined joint axes through vectors

$$\mathbf{v}_{\text{knee}} = \mathbf{p}_{\text{lateral}} - \mathbf{p}_{\text{medial}}, \quad (1)$$

where $\mathbf{p}$ denotes marker positions. These axes were enforced in the Vicon pipeline to define the local coordinate systems.

Participants also performed five right-leg swings before each walking trial to allow accurate synchronization between IMU sensors and the other devices.

C. Vicon Preprocessing: Gap Filling and Labelling

1) *Gap Filling in Vicon Nexus*: The Vicon motion capture system detects reflective markers and reconstructs body segments. When markers are occluded by the participant's arms, clothing, or the treadmill structure, missing data (gaps) occur. These gaps were corrected using Vicon Nexus gap-filling algorithms. Table I summarizes the available methods.

2) *Labelling and Model Output Adjustment*: Markers were labelled using a standard lower-limb model in Vicon Nexus. The local coordinate system of each segment was aligned with anatomically meaningful axes. For example, the knee flexion/extension axis was defined along the vector connecting lateral and medial knee markers. Joint angles were then computed by expressing distal segments relative to proximal segments in these local coordinates.

D. IMU–Vicon Synchronization

IMU and Vicon systems were recorded with independent clocks. Let $s_{\text{IMU}}(t)$ denote an IMU signal (e.g., thigh angular velocity) and $s_{\text{Vicon}}(t)$ the corresponding quantity obtained from differentiated Vicon joint angles. Synchronization was performed by estimating a time shift τ^* that minimizes the ℓ_2 norm between the two:

$$\tau^* = \arg \min_{\tau} \|s_{\text{IMU}}(t + \tau) - s_{\text{Vicon}}(t)\|_2. \quad (2)$$

In practice, this was implemented by aligning the prominent spikes in the right-leg angular velocity signals obtained during five right-leg swings at the start of each trial recording. Using MATLAB, the IMU time axis was shifted by τ^* so that both signals shared consistent temporal landmarks. Figures 4 and 5 in the Results section illustrate this alignment.

E. EMG Preprocessing and Automated Disconnection Correction

The EMG preprocessing component is the most algorithmically involved part of the methodology. The implementation uses a sequence of steps: file validation, first-level flag generation, segment detection with dual thresholds, slope-based boundary refinement, spectrum-based noise synthesis, and batch processing for multiple subjects and conditions. This subsection formalizes each step.

1) *EMG File Structure and Validation*: Raw EMG signals were stored in comma-separated value (CSV) files with a specific structure:

- The 4th line contained the column headers.
- Numeric data rows began at line 6.

Let L be the total number of lines in a file and let the header vector be

$$\mathbf{h} = [h_1, h_2, \dots, h_C],$$

where C is the expected number of columns. The file validation enforced:

- 1) No completely empty rows in the middle of the file. Let i^* be the index of the last non-empty line. Any empty line $i < i^*$ triggered rejection.
- 2) No line beyond the header with more than C comma-separated entries. If a line contained $C_i > C$ cells, the file was considered corrupted.

If either condition was violated, the file was rejected from automated processing.

2) *First-Level Thresholding and Flag Generation*: For each EMG channel labelled as an IM EMG channel (columns whose header string contained “IM EMG”), the raw signal was converted to a numeric array

$$\mathbf{x} = [x_1, x_2, \dots, x_N],$$

where N is the number of samples. Non-numeric or non-finite entries were coerced to NaN and ignored where appropriate.

A first-level amplitude threshold

$$T_1 = 5 \times 10^{-7}$$

was applied to detect near-zero segments that could correspond to disconnections. A binary flag sequence

$$\mathbf{f} = [f_1, f_2, \dots, f_N]$$

was constructed as

$$f_i = \begin{cases} 1, & \text{if } |x_i| < T_1, \\ 0, & \text{otherwise.} \end{cases} \quad (3)$$

Channels whose finite values were all zero, i.e.,

$$\{i \mid x_i \text{ finite}\} \neq \emptyset, \quad \text{and } x_i = 0 \text{ for all finite } i,$$

were treated as “sensors not used” and excluded from further processing.

The result of this step is a DataFrame of flag columns named “<IM EMG column> flag”, each encoding first-level low-amplitude regions.

3) *Detection of Candidate Disconnection Runs*: Given the flag vector $\mathbf{f}$, contiguous runs of ones were interpreted as candidate disconnection regions. Formally, a run is an index interval

$$[s, e) = \{s, s+1, \dots, e-1\}$$

such that

$$f_i = 1 \text{ for } i \in [s, e), \quad f_{s-1} = 0 \text{ (or } s = 1), \quad f_e = 0 \text{ (or } e = N+1). \quad (4)$$

Only runs with length

$$d = e - s \geq \text{min_run_len} = 20$$

samples were retained as candidate gaps.

4) *Second-Level Threshold Expansion*: A second, more permissive amplitude threshold

$$T_2 = 5 \times 10^{-6}$$

was used to expand each candidate run to include its low-amplitude neighborhood. For an initial candidate run $[s, e)$, the expanded segment $[s_2, e_2)$ is defined as:

$$s_2 = \max \{i \leq s \mid i = 0 \text{ or } |x_i| > T_2\} + 1, \quad (5)$$

$$e_2 = \min \{j \geq e \mid j = N \text{ or } |x_j| > T_2\}. \quad (6)$$

Intuitively, the disconnection segment grows backward and forward until the signal magnitude exceeds T_2 , which provides a buffer around the strictly low-amplitude region identified by the first threshold.

5) *Slope Computation and Slope-Based Boundary Refinement*: To avoid abrupt changes at the boundaries of each corrected segment, slope information was used to refine the start and end indices of candidate gaps. First, a finite-difference estimate of the slope was computed from the raw EMG signal:

$$d_i = x_{i+1} - x_i, \quad i = 1, \dots, N-1, \quad (7)$$

where x_i denotes the EMG amplitude at sample i and N is the total number of samples.

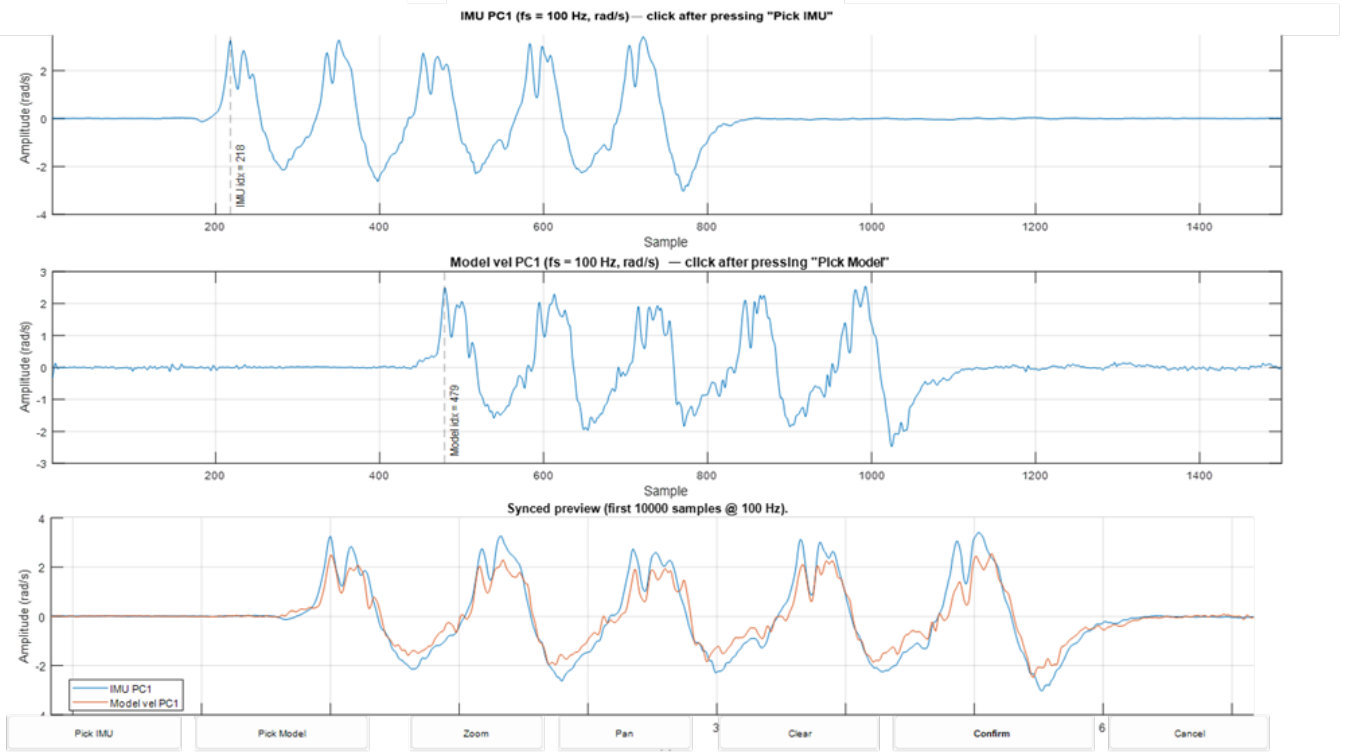


Fig. 4. IMU synchronization: PCA-filtered thigh IMU gyroscope data aligned with Vicon-derived hip angular velocity.

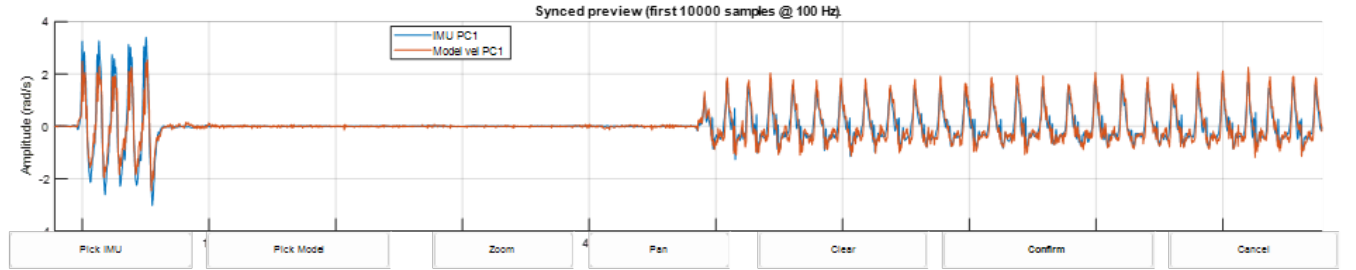


Fig. 5. IMU synchronization—full frame view showing consistent timing of key gait events.

Because finite differences amplify small frame-to-frame fluctuations, the raw slope sequence $\mathbf{d}$ typically contains high-frequency noise. To obtain a smoother and more reliable representation of the local trend, $\mathbf{d}$ was convolved with a moving-average kernel of length w :

$$k_j = \frac{1}{w}, \quad j = 1, \dots, w, \quad (8)$$

and

$$s_i^{(m)} = (\mathbf{d} * \mathbf{k})_i = \sum_{j=1}^w k_j d_{i-j+1}, \quad (9)$$

for all indices i where the sum is well-defined. The superscript (m) in $s_i^{(m)}$ denotes that this is the *smoothed* (or *mean-filtered*) slope, distinguishing it from the raw finite-difference slope d_i .

This convolution acts as a low-pass filter on the slope signal: it suppresses isolated spikes, sensor noise, and single-sample irregularities while preserving the meaningful upward

or downward trends around each gap boundary. Using the smoothed slope sequence $s^{(m)}$ ensures that boundary refinement decisions respond only to consistent slope patterns rather than noise-induced fluctuations. If this smoothing step were omitted, the refinement procedure would become highly sensitive to jitter in the raw EMG signal, causing unstable or oscillatory boundary adjustments and ultimately producing visually discontinuous or physiologically implausible corrected EMG segments.

For each expanded segment $[s_2, e_2)$, the boundary indices were further refined using the sign and stability of the smoothed slopes with a slope stability tolerance $\varepsilon = \text{slope_stability_tol}$. Conceptually:

- If the slope near the start boundary is consistently negative (or positive) and its magnitude is larger than ε , the start index is adjusted backward or forward until the slope magnitude no longer exceeds ε .

- A similar rule is applied to the end boundary, but scanning in the opposite direction.

After refinement, each candidate disconnection region is represented as

$$[r_s, r_e), \quad \text{where } r_s, r_e \in \{1, \dots, N\}, \quad r_s < r_e.$$

A small offset `forceBack` was also applied to shift the start boundary slightly backward (when possible) to ensure a conservative inclusion of possibly corrupted samples.

6) *Global Segment Length Constraint and File Rejection:* Segments with duration

$$d = r_e - r_s$$

were evaluated against a maximum gap length

$$d \geq \text{max_gap_len} = 1000 \quad \Rightarrow \quad \text{segment too long.}$$

If any channel exhibited at least one segment with $d \geq \text{max_gap_len}$ that was not explicitly allowed by pre-specified exception lists, the entire file was rejected from automated correction. This prevented the algorithm from filling excessively long gaps that are unlikely to be recoverable in a realistic way.

7) *Context Windows and Exclusion of Overlapping Segments:* For each accepted segment $[r_s, r_e)$ with length $d = r_e - r_s$, context windows of length d were defined before and after the gap:

$$\mathcal{B} = \{\max(1, r_s - d), \dots, r_s - 1\}, \quad (10)$$

$$\mathcal{A} = \{r_e, \dots, \min(N, r_e + d - 1)\}. \quad (11)$$

To avoid contamination from other correction segments, indices belonging to any disconnection region were removed from these sets, resulting in usable index sets

$$\mathcal{B}_{\text{usable}} \subseteq \mathcal{B}, \quad \mathcal{A}_{\text{usable}} \subseteq \mathcal{A}. \quad (12)$$

If both usable sets were empty, the segment was skipped (no correction applied), because there was insufficient valid context to estimate the local signal statistics. If only one side was non-empty, the available side was used for both before and after statistics.

8) *Context Statistics and Linear Bridge:* From the usable context sets, local means and standard deviations were computed:

$$\mu_B = \text{mean}\{x_i : i \in \mathcal{B}_{\text{usable}}\}, \quad \sigma_B = \text{std}\{x_i : i \in \mathcal{B}_{\text{usable}}\}, \quad (13)$$

$$\mu_A = \text{mean}\{x_i : i \in \mathcal{A}_{\text{usable}}\}, \quad \sigma_A = \text{std}\{x_i : i \in \mathcal{A}_{\text{usable}}\}. \quad (14)$$

If μ_B (or μ_A) was undefined, it was replaced by the other side's mean.

A deterministic linear “bridge” across the gap was defined using boundary values:

$$y_{\text{start}} = \begin{cases} x_{r_s-1}, & r_s > 1 \text{ and } x_{r_s-1} \text{ finite,} \\ \mu_B, & \text{otherwise,} \end{cases} \quad (15)$$

$$y_{\text{end}} = \begin{cases} x_{r_e}, & r_e \leq N \text{ and } x_{r_e} \text{ finite,} \\ \mu_A, & \text{otherwise.} \end{cases} \quad (16)$$

For $k = 0, 1, \dots, d-1$, the interpolation is

$$\ell_k = y_{\text{start}} + \frac{k}{d-1} (y_{\text{end}} - y_{\text{start}}), \quad (17)$$

yielding a line $\ell = [\ell_0, \dots, \ell_{d-1}]$.

Time-varying mean and standard deviation trajectories across the gap were also defined:

$$\mu_k = \mu_B + \frac{k}{d-1} (\mu_A - \mu_B), \quad (18)$$

$$\sigma_k = \max\left(\sigma_{\min}, \sigma_B + \frac{k}{d-1} (\sigma_A - \sigma_B)\right), \quad (19)$$

where $\sigma_{\min} > 0$ is a small constant (e.g., 10^{-12}) to avoid numerical degeneracy.

9) *Spectrum-Based Colored Noise Synthesis:* To preserve the frequency characteristics of the EMG signal, colored noise was synthesized using the amplitude spectra of the context windows. Let

$$\mathbf{v}_B = [x_i - \mu_B : i \in \mathcal{B}_{\text{usable}}], \quad \mathbf{v}_A = [x_i - \mu_A : i \in \mathcal{A}_{\text{usable}}],$$

and let $\mathbf{w}$ be a Hann window of appropriate length. The windowed signals are

$$\tilde{\mathbf{v}}_B = \mathbf{v}_B \odot \mathbf{w}, \quad \tilde{\mathbf{v}}_A = \mathbf{v}_A \odot \mathbf{w}, \quad (20)$$

where $\odot$ denotes element-wise multiplication. Their discrete Fourier transforms (real FFT) are

$$\mathbf{S}_B = \text{rFFT}(\tilde{\mathbf{v}}_B), \quad \mathbf{S}_A = \text{rFFT}(\tilde{\mathbf{v}}_A). \quad (21)$$

The corresponding amplitude spectra are

$$\mathbf{A}_B = |\mathbf{S}_B|, \quad \mathbf{A}_A = |\mathbf{S}_A|. \quad (22)$$

When both sides are available, a blended amplitude spectrum is formed as

$$\mathbf{A} = \frac{1}{2} (\mathbf{A}_B + \mathbf{A}_A) \quad (23)$$

(after truncation to the minimum length if necessary). If only one side is available, that side's amplitude spectrum is used; if neither is available, the algorithm falls back to white noise.

Random phases $\phi \sim \mathcal{U}(0, 2\pi)$ are drawn independently for each frequency bin to form a complex spectrum

$$\mathbf{S}_{\text{target}} = \mathbf{A} \odot e^{j\phi}, \quad (24)$$

and a real-valued colored noise signal of length d is obtained via the inverse real FFT:

$$\mathbf{n} = \text{irFFT}(\mathbf{S}_{\text{target}}, n = d), \quad (25)$$

which, in component form, can be written as

$$n_t = \frac{1}{d} \sum_{k=0}^{d-1} S_{\text{full}}[k] e^{j2\pi kt/d}, \quad t = 0, 1, \dots, d-1, \quad (26)$$

where S_{full} is the Hermitian-symmetric extension of the one-sided spectrum $\mathbf{S}_{\text{target}}$ returned by `rFFT` (i.e., the negative-frequency bins are reconstructed as complex conjugates of

the positive-frequency bins), ensuring that the time-domain sequence $\mathbf{n}$ is real-valued. The noise is then normalized to zero mean and unit variance:

$$\tilde{\mathbf{n}} = \frac{\mathbf{n} - \text{mean}(\mathbf{n})}{\text{std}(\mathbf{n})}. \quad (27)$$

10) *Final Artificial Signal Within the Gap*: The final artificial EMG signal $\mathbf{x}^{\text{gap}}$ within the gap is constructed by scaling the colored noise by the time-varying standard deviation and adding it to the linear bridge:

$$x_k^{\text{gap}} = \ell_k + \sigma_k \tilde{n}_k, \quad k = 0, \dots, d-1. \quad (28)$$

The original samples $x_{r_s}, \dots, x_{r_e-1}$ are then replaced with $x_0^{\text{gap}}, \dots, x_{d-1}^{\text{gap}}$. By construction, the added noise has zero mean, and its standard deviation follows the local trend estimated from the surrounding valid segments. This ensures that the artificial segment blends smoothly into the original EMG signal while preserving realistic variability and spectral content.

11) *Per-Channel Reports and Segment Tables*: For each channel, the algorithm records:

- The number of corrected segments.
- The start index, end index, and duration for each corrected segment.

These records are assembled into a table of the form

$$\{\text{file, channel, start, end, duration}\}, \quad (29)$$

which can later be exported as a summary DataFrame for quality control and documentation.

Channels or segments that were skipped (e.g., due to insufficient usable context or all-zero sensors) are also logged with textual reasons.

12) *Writing Corrected Files*: The original EMG CSV files included metadata lines (e.g., acquisition information) preceding the data. After correction, the metadata lines are preserved, and the corrected numerical data are written to a new output file with a suffix such as “_corrected”. The corrected files have the same shape (same number of rows and columns) as the original data section, enabling them to be used as drop-in replacements for downstream processing, including neural network training.

13) *Visualization of Corrected Segments*: For visualization and manual verification, context plots around each corrected segment are generated. For each segment with duration d exceeding a specified threshold (e.g., $d > 100$ samples), a window of $\pm \text{plotWindow}$ samples around the corrected region is displayed:

$$[\max(1, r_s - \text{plotWindow}), \min(N, r_e + \text{plotWindow})]. \quad (30)$$

Within this window, the original and corrected signals are overlaid, with vertical lines marking the start and end of the corrected interval. These plots are saved with filenames encoding the EMG channel, segment index, and duration, and organized in a subject- and condition-specific directory structure.

F. Batch Processing Across Subjects and Conditions

The complete algorithm supports batch processing of multiple subjects and experimental conditions. For each file, the batch process:

- 1) Validates the file and constructs first-level flags.
- 2) Runs the full correction algorithm with the dual-threshold and slope-based logic.
- 3) Rejects files with excessively long segments or structural corruption.
- 4) Generates correction plots for all accepted segments.
- 5) Records summary statistics such as the number of corrected channels and segments.

A final batch-level report collects:

- A list of successfully processed files and the number of corrections applied.
- A list of “bad files” (e.g., missing input, structural errors, or long-segment rejections) along with reasons and, if applicable, the offending channel and segment indices.

This batch-oriented design ensures that the preprocessing pipeline can scale to large datasets and that problematic files are isolated and documented without interrupting the processing of other data.

IV. RESULTS

A. Artificial EMG Data Generation

Frequent EMG disconnections were observed across all participants, particularly for sensors #3, #7, and #10. To address this, the dual-threshold and spectrum-based reconstruction algorithm described in the Methods section was applied.

The first-level threshold (T_1) identified near-zero regions. The second-level threshold (T_2) and slope-based refinement expanded and smoothed the boundaries. Within each refined gap, a linear bridge was combined with colored noise whose spectrum was estimated from windows before and after the gap. This approach produced continuous EMG signals that preserved the local variance and frequency characteristics of the original recordings.

Figure 6 illustrates an example comparison between the original disconnected signal and the reconstructed version. A zoomed-in view of one corrected segment is shown in Fig. 7, highlighting how the artificial signal blends smoothly with the surrounding EMG activity. All the raw EMG recordings went through this process, creating a suitable EMG dataset for the neural network model training process next term.

B. IMU Synchronization

Figure 4 shows the gyroscope signal from the right thigh IMU with principal component analysis (PCA) applied, and Figure 5 shows the corresponding right hip angular velocity obtained by differentiating the Vicon hip rotation data. After alignment, the two signals exhibited closely matching peaks and troughs, confirming successful synchronization between the IMU and Vicon motion capture systems. This step ensured that the IMU data could be directly compared with and integrated into the rest of the data captured by Vicon for subsequent analysis.

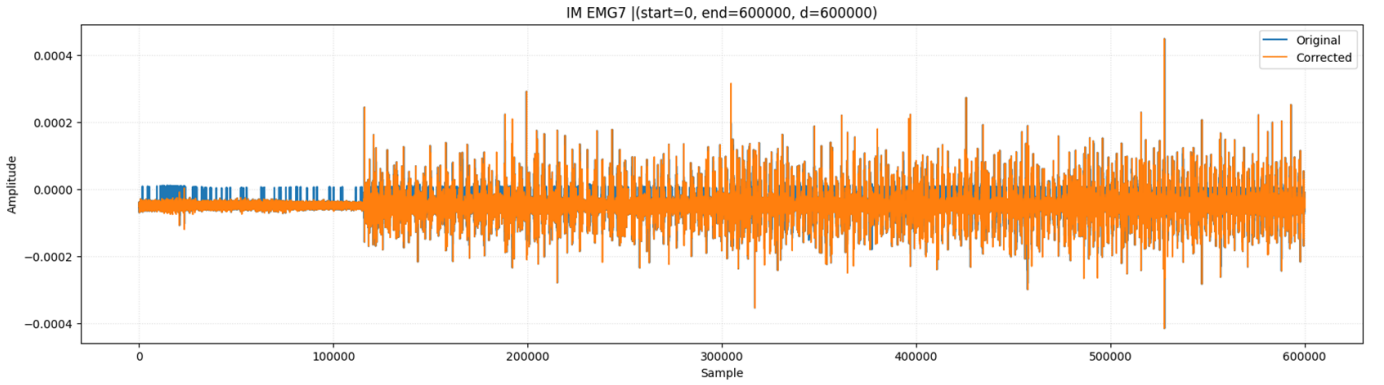


Fig. 6. Comparison of original disconnected EMG signals vs. artificially filled EMG signals.

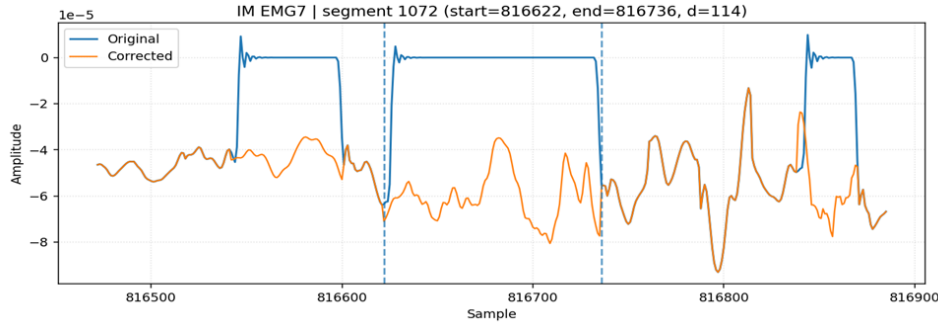


Fig. 7. Zoomed comparison of original vs. artificial EMG signals in a corrected segment.

V. DISCUSSION

As outlined in the Results, one major challenge faced during data collection was EMG disconnection issues. These disconnections led to unusable data segments if left uncorrected. The solution was to replace the disconnected regions with a combination of linear interpolation and spectrally shaped noise.

Writing the Python script for this solution was the most challenging part, largely because the data files were heterogeneous and the logic had to be robust to a variety of corner cases, including:

- Corrupted rows and unexpected empty lines in EMG files.
- All-zero channels associated with unused or malfunctioning sensors.
- Extremely long candidate gaps that required rejection rather than correction.

The final pipeline is modular: the flag generation, segment detection, slope refinement, spectral analysis, and batch processing are all implemented as separate components. This design will make it easier to adapt the algorithm to other muscles, sampling rates, or recording setups.

VI. CONCLUSION

This term, the main focus was on collecting gait cycle data and working through preprocessing steps, such as gap filling, labelling, synchronization, and EMG data cleaning. Although

challenges came up, especially with EMG disconnections, solving them with mathematical concepts gave me hands-on experience in signal processing and practical insight into the complexities of working with human movement data.

The progress made this term provides a solid base for the Fall term, where neural network training will be the main focus. Training the model to generate a full future gait cycle for joint trajectories will bring the project closer to its goal of supporting rehabilitation through exoskeleton technology.

REFERENCES

- [1] W. H. Organization, “Mortality and global health estimates,” <https://www.who.int/data/gho/data/themes/mortality-and-global-health-estimates>, 2025, accessed 18 September 2025.
- [2] —, “Spinal cord injury,” <https://www.who.int/news-room/fact-sheets/detail/spinal-cord-injury>, 2024, accessed: Sep. 18, 2025.
- [3] R. P. Price and N. L. P. Choy, “Investigating the relationship of the functional gait assessment to spatiotemporal parameters of gait and quality of life in individuals with stroke,” *Journal of Geriatric Physical Therapy*, vol. 42, no. 4, pp. 256–264, Oct-Dec 2019.
- [4] A. Suebsomran, N. Manoch, and P. Kwanthong, “Development and control of a lower limb exoskeleton robot,” in *2022 International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME)*. Maldives: IEEE, 2022, pp. 1–6.